SCE331 Data Mining: Major Algorithms by Topic

1. Frequent Itemsets and Association Rules

- Apriori (A-Priori).
- PCY (Park–Chen–Yu).
- SON
- Toivonen's algorithm.
- Association-rule generation with confidence-based pruning.

2. Similar-Item Search

- Shingling and hashed shingling.
- MinHash.
- Locality-Sensitive Hashing (LSH), including MinHash signature banding.

3. Clustering (counts as only one algorithm)

- Agglomerative hierarchical clustering.
- k-means.
- BFR
- CURE.

4. Recommender Systems

- Content-based filtering.
- User-based and item-based collaborative filtering.
- Baseline-adjusted collaborative filtering.
- Regularized latent-factor matrix factorization, including user/item biases.

5. Dimensionality Reduction and Transforms (counts as only one algorithm)

- Principal Component Analysis (PCA), including PCA via SVD.
- Singular Value Decomposition (SVD) and truncated SVD.

6. Link Analysis and Graph Proximity

- PageRank, including teleportation and dead-end handling, power iteration and sparse matrix–vector computation.
- Topic-specific / personalized PageRank.
- Random Walk with Restart (RWR): single-seed personalization version
- TrustRank: trust propagation from trusted seed pages.

7. Community Detection and Graph Partitioning

- Girvan–Newman: divisive clustering by repeated edge-betweenness removal.
- Louvain: greedy modularity optimization with community aggregation.
- Spectral graph partitioning using the Laplacian and Fiedler vector.
- k-way spectral clustering: recursive bipartitioning or multiple-eigenvector embedding.
- Personalized-PageRank-based local clustering with a conductance sweep.

8. Sampling (counts as only one algorithm)

- Uniform sampling with/without replacement; biased and stratified sampling.
- Fixed-proportion element sampling and hash-based key sampling.
- Reservoir sampling for a fixed-size stream sample.

9. Stream Mining

- Hash-table membership testing and single-hash bit-array filtering as baselines.
- Bloom filtering for approximate membership testing.
- Flajolet–Martin for distinct-count estimation.
- AMS (Alon–Matias–Szegedy) for frequency-moment estimation; marked advanced in the stream lecture.
- Exponentially decaying-window counting for recent frequent items/itemsets.

11. Supporting Preprocessing Methods (Does not count as a major algorithm)

- TF–IDF weighting.
- Equi-width, equi-log, and equi-depth discretization.
- Binary / one-hot encoding.
- Standardization and min–max scaling.
- Missing-record deletion and imputation.
- Distance-threshold neighborhood-graph construction and heat-kernel weighting.
- Supervised/unsupervised feature selection: introduced as approaches, without a named selection algorithm.